

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Sr. Ali Babu (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sr. Ali Babu (192524038)

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20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Descriptive Written Test

①

Smart Home Automatic System

A Smart Home Automation System is being to manage lighting, Temperature, and security by taking information from the motion detection, Temperature changes. The Artificial Intelligence enables the Intelligence decision from the Environment.

Types of Intelligent Agents

1. Simple Reflex Agent :-

A Simple Reflex agent makes Decision Based on the Current Input. It Does not remember about previous.

Example

1. If Motion is detected → Turn On the Lights
2. If No motion is detected → Turn off the Lights.

Advantage :-

1. Easy fast response
2. Easy to Design.

2. Model Base Reflex Agent :

It Maintains an Internal Model of the Environment.

User Can Takes Decision By Current and Previous Information.

Example :

- Rember that the Bed room Light is On.
- Detects that the window is open while AC is running.

Advantages :

- Better Decision making.
- Uses memory of previous Tasks.
- More Complex previous Simple Reflex agent.

3. Goal Based Agent :

- Make Decision Based on acheiving a Specific Goal.

- Searching Techniques By User.

Example :

- * Maintain the room Temperature 24°C
- * Ensure all Doors are Locked.

Advantage :

- To a Intelligent planning.
- Can Evaluate different actions.

4. Utility Based Agent :

Select the action that provide the highest Benefit.

Example $\Rightarrow$ *

- * User Comfort
- * Security
- * Electricity Cost.

Advantage :

* It provides optimal Decision.

Best Agent

- * Model Based reflex agent
User takes Decision on Current and previous Information.
- * Goal Based Agent
acheive Objectives Such as maintaining Temperature.
- * Utility Based Agent
Optimizing Comfort & Energy.

Conclusion

The Smart Home automation System is most effective for Ensuring Comfort, Energy, Efficiency & Security.

2. Robot Navigation in an Unknown Maze

A Robot is placed in an unknown maze and must reach without knowing the maze layout beforehand. The robot must explore the environment & determine the Best path using AI.

1. Uniform Cost Search.

→ Uniform Cost Search expands the node with the lowest path cost.

Working:

Start Initial position

↓
Store needs in priority Queue.

↓
Expand node with Smallest path cost

↓
Continuous until goal reach

Advantage

1. find least cost path.
2. Optimal solutions.

Disadvantage

1. High Memory Usage.
2. Slow for Every Large maze

(2)

Interactive Deepening Search

IDS Combines the advantages of Depth first Search and Breadth first Search.

Working

Search with depth limit 0.

↓
Increase Depth limit to 1

↓
Then 2 So on

↓
Repeat until goal reach.

Advantage

1. Very Low memory required
2. Complete.

Complexity

Time: $O(b^d)$

Space: $O(bd)$

b - Branching factor

$d \rightarrow$ Depth of Solution.

3. Intelligent Agents in maze Navigation

- $\rightarrow$ Stores visited location.
- $\rightarrow$ affects Decision making.

Conclusion

Goal Based agent using provides the most reliable & optimal Navigation for an unknown maze.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand